

HW05 - AI-Assisted Performance Testing

Student ID: 23127124

SUT: EShop

Tool: Apache JMeter 5.6.3

Test date: 2026-09-03

Base URL: `http://127.0.0.1:3000`

Public repository: <https://github.com/ngothoaidev/23127124-HW05-Performance-Testing>

Demo video: <https://youtu.be/kmkiT8y2rXQ>

1. Executive summary

This assignment evaluates the EShop backend with one repeatable, data-driven business process under Load, Stress, Spike, and Soak conditions. The selected features are FR-01 Registration, FR-07 Shopping Cart, and FR-17 Coupon Management. Supporting login, product-read, coupon-application, and checkout requests are included to satisfy the required auth-heavy, read-heavy, and transactional endpoint groups.

The measured Load run sustained 120.94 HTTP RPS with zero failed samples and p95 of 23 ms. A first Stress level at 80 VUs reached 523.07 RPS with p95 of 20 ms. At 400 VUs, throughput rose only to 849.86 RPS while p95 increased to 636 ms, demonstrating a latency saturation region. The corrected 600-VU Spike reached 904.16 RPS with p95 of 914 ms, p99 of 1,146 ms, and 0.1658% errors. The ten-minute Soak sustained 926 RPS at 250 VUs with zero errors and p95 of 311 ms, establishing the measured endurance threshold for this local hardware configuration.

2. Environment and reproducibility

Item	Configuration
Operating system	Windows 11 Home 10.0.26200
Hostname	THOAI
CPU	AMD Ryzen 7 8845H, 16 logical processors
RAM	16 GB installed; 13.81 GB usable during the original measurement
Backend	Node.js 24.18.0, Express 5.2.1, SQLite
Java	Eclipse Temurin 21.0.12.1 LTS
JMeter	5.6.3
Deployment	Backend and load generator on the same host

Only the backend is started during measurement. This avoids frontend overhead and uses loopback networking to reduce unrelated network variance. The limitation is that JMeter and the SUT compete for the same CPU and RAM, so the results describe the endurance threshold of this complete local setup rather than an independently hosted production server.

`server.js` imports `database.js`, which recreates and seeds SQLite at backend startup. The backend is therefore restarted before each measured run. This also clears the process-local cart object.

3. AI-assisted design and human review

3.1 Workflow

1. Setup administrator login and creation of a unique coupon (FR-17).
2. Unique data-driven user registration (FR-01).
3. User login and JWT correlation.
4. Product list and detail reads.
5. Add product to cart and view cart (FR-07).
6. Apply the run coupon and complete checkout.
7. Teardown administrator login and deletion of the run coupon (FR-17).

The CSV contains 1,000 unique virtual-user identities and parameterized product/order data. Tokens, user IDs, coupon IDs, and order IDs are extracted from JSON responses.

3.2 Corrections made after executing AI output

- Replaced an unrealistic single-role interpretation with explicit administrator setup/teardown and a customer main group.
- Added product reads because the selected features alone did not meet the read-heavy requirement.
- Changed think-time from a Thread Group timer applied before every request to one pause per iteration.
- Increased CSV cardinality from 250 to 1,000 for the 600-VU Spike.
- Guarded the business flow so failed token extraction cannot cause cascading 403 errors.
- Preserved the rejected calibration results as audit evidence.

4. Test profiles

Scenario	VUs	Ramp-up	Duration	Report view
Load	15	30 s	180 s	Summary Report
Stress calibration	80	120 s	240 s	Aggregate Report
Stress saturation	400	60 s	180 s	Aggregate Report
Spike	600	2 s	90 s	Aggregate Graph
Soak	250	60 s	600 s	HTML Dashboard/raw JTL

JMeter was executed in non-GUI mode. Raw JTL files are authoritative; Transaction Controller parent samples are excluded from HTTP request counts and throughput calculations.

5. Results

Run	HTTP samples	Error rate	RPS	Average	p95	p99	Max	Max backend working set
Load 15 VU	21,929	0%	120.94	3.72 ms	23 ms	28 ms	144 ms	177.68 MB
Stress 80 VU	126,316	0%	523.07	5.50 ms	20 ms	38 ms	290 ms	183.77 MB
Stress 400 VU	154,758	0%	849.86	280.66 ms	636 ms	770 ms	1,044 ms	197.57 MB
Spike 600 VU	83,242	0.1658%	904.16	467.33 ms	914 ms	1,146 ms	2,197 ms	210.68 MB
Soak 250 VU	557,131	0%	926.00	146.90 ms	311 ms	396 ms	2,466 ms	211.34 MB

5.1 Load

The Load run passed the proposed p95 < 1,000 ms and error-rate < 1% criteria by a wide margin. It establishes a low-contention baseline rather than the maximum system capacity.

5.2 Stress

Increasing concurrency from 80 to 400 VUs (5x) increased throughput only from 523.07 to 849.86 RPS (1.62x), while p95 rose from 20 to 636 ms. This flattening of throughput combined with a sharp latency rise indicates saturation even without HTTP errors.

5.3 Spike

The corrected Spike run recorded 69 refused registration connections, 67 refused login connections, and two 401 login responses. The 0.1658% error rate remained below 1%, but p99 exceeded the one-second target. The service degraded under the abrupt load but completed the run without a process crash.

5.4 Endurance threshold

The 250-VU Soak sustained 926.00 RPS for ten minutes with 557,131 HTTP samples, zero errors, average latency 146.90 ms, p95 311 ms, and p99 396 ms. The maximum observed backend working set was 211.34 MB. This is the maximum stable sustained rate actually demonstrated on this hardware; it is not a universal capacity claim.

Memory did not form a clear flat plateau. Average working set was 192.74 MB in minute 5, 194.76 MB in minute 6, 197.89 MB in minute 7, 205.91 MB in minute 8, and 207.07 MB in minute 9. Source review confirms that checkout never clears the process-local cart and every cart POST appends another object. The result therefore supports an application-level unbounded-retention issue. Longer observation would be required to separate the retained-cart growth rate from Node.js garbage-collection and growing cart-response overhead.

6. AI analysis and misinterpretation hunt

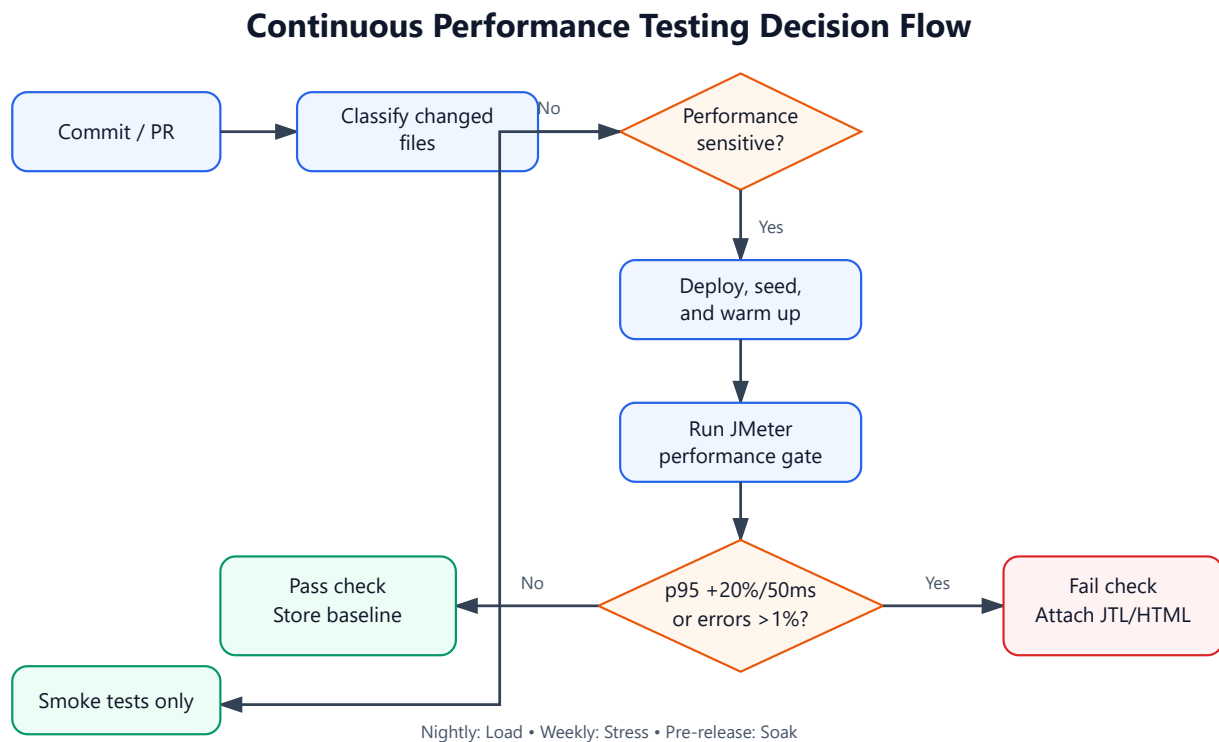
The preserved AI analysis and human review are included under [analysis/](#). The main corrections are that a short Stress run cannot prove an endurance threshold, a low aggregate Spike error rate does not erase connection refusals and p99 regression, and short-run memory growth does not by itself prove a leak. Recommendations were checked against the actual Node.js/SQLite implementation.

7. Genuine issues

Two verified issues were published with JMeter dashboard screenshots:

1. Connection refusals and p99 regression during a corrected 600-VU Spike.
2. Checkout does not clear the process-local cart, creating functional and sustained-memory risk.

8. Continuous Performance Testing proposal



The proposed pipeline classifies changed files and runs a performance gate for backend routes, database changes, authentication, serialization, dependencies, or runtime configuration. It compares p95/error rate/RPS with the median of five valid runs on equivalent hardware. A regression fails when p95 increases by more than both 20% and 50 ms, or when error rate exceeds 1%. Full Load runs execute nightly, Stress weekly, and Soak before releases.

The full flowchart and trade-off analysis are in [docs/continuous-performance-testing.md](#).

9. AI critique

The AI was useful for quickly converting the assignment into an executable JMeter design, but its first outputs were incomplete in three important ways. First, it treated FR-01, FR-07, and FR-17 as a natural single-user journey, even though coupon management is an administrator operation and the selected features do not provide a suitable read-heavy flow. Human review corrected this by separating coupon setup and teardown from the customer workflow and adding product reads. Second, the generated JMeter plan placed a random timer at Thread Group scope. JMeter therefore applied the delay before every sampler instead of once per business iteration, reducing a ten-second smoke run from 142 to only 26 samples. Third, the initial CSV contained 250 identities while the Spike test used 600 virtual users. Recycled identities caused duplicate registrations, and failed authentication produced thousands of

misleading 403 responses. The corrected plan uses 1,000 rows and prevents the business flow from running without a successfully extracted token. The AI also overinterpreted short-run memory growth as proof of a leak and suggested a conventional database connection pool even though the SUT uses embedded SQLite. These failures occurred because the model reasoned from generic performance-testing patterns without executing JMeter scope semantics, checking data cardinality against peak concurrency, or validating the actual database architecture. The main lesson is that AI output should be treated as a testable hypothesis. Collaboration is reliable only when every design decision is exercised with a small run, raw logs are preserved, derived metrics are independently recalculated, and recommendations are checked against the implementation before acceptance.

10. Optimization assessment

Recommendation	Decision	Reason
Enable SQLite WAL	Feasible; benchmark first	It can improve read/write concurrency but does not remove every serialized write limit.
Move carts to Redis	Feasible	It removes process-local retention and enables multi-process deployment, at added operational cost.
Cache product lists	Feasible with invalidation	The endpoint is read-heavy; product CRUD must invalidate cached values.
Ordinary product-name index	Unsupported	The current <code>LIKE '%keyword%'</code> query has a leading wildcard and generally cannot use a normal B-tree efficiently.
Database connection pool	Hallucinated for this architecture	SQLite is embedded rather than a network database that benefits from conventional connection pooling.
Enable Node.js cluster immediately	Unsafe as stated	Process-local carts must be externalized or consistently routed first.

11. Conclusion

The measured local SUT scales efficiently at low concurrency but enters a latency-saturation region before 400 VUs. At 600 abruptly started VUs, throughput rises modestly while p99 passes one second and the listener observes connection refusals. The demonstrated endurance threshold is 926 RPS at 250 VUs for ten minutes, with p95 311 ms, 0% errors, and a 211.34 MB observed memory ceiling. The upward memory trend and cart implementation justify a verified functional/performance issue, not an unsupported generic memory-leak claim.

12. Student-captured evidence and verification reruns

The student captured the completed JMeter command beside Task Manager for each required workload and supplied a `dxdiag` System screenshot. The three verification reruns were independently parsed from their raw JTL files:

Rerun	HTTP samples	Errors	RPS	Average	p95	p99	Max working set
Load, 15 VUs	22,583	0	124.66	1.95 ms	6 ms	24 ms	177.10 MB
Stress, 400 VUs	233,905	0	1,290.52	149.28 ms	349 ms	453 ms	201.43 MB
Spike, 600 VUs	111,647	0	1,218.19	376.18 ms	743 ms	865 ms	212.54 MB

These reruns confirm that the test plans completed with their declared concurrency. Their better error rates also demonstrate run-to-run variation on a shared workstation; they do not invalidate the earlier recorded Spike connection-refusal event.

12.1 Load evidence

The screenshot displays a Windows PowerShell window and the Windows Task Manager, illustrating system resource usage during a load test.

Windows PowerShell Output:

```
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/stop-backend.ps1
Stopped backend PID 36856.
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/start-backend.ps1 -Scenario EvidenceLoad
Backend PID 23504 ready at http://127.0.0.1:3000; database reseeded on startup.
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/run-jmeter.ps1 -Scenario Load
Completed Load: threads=15 ramp=30s duration=180s
JTL: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Load-20260903200804\23127124_Load_20260903.jtl
HTML: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Load-20260903200804\html
Resources: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Load-20260903200804\resource-usage.csv
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5>
```

Task Manager Processes:

Name	Status	CPU	Memory	Disk	Network
Open/DK Platform binary		15.1%	241.3 MB	1.7 MB/s	0 M
Microsoft OneDrive Sync Serv...		14.3%	68.7 MB	0.4 MB/s	0 M
Microsoft OneDrive		7.0%	407.1 MB	0.3 MB/s	0 M
Codex (10)		1.7%	962.4 MB	0.1 MB/s	0 M
Task Manager		0.9%	80.4 MB	0 MB/s	0 M
Antimalware Service Executable		0.9%	186.1 MB	0.1 MB/s	0 M
Spotify (7)		0.5%	184.7 MB	0 MB/s	0 M
System		0.4%	0.1 MB	1.7 MB/s	0 M
Service Host: Network Service		0.4%	5.2 MB	0 MB/s	0 M
Service Host: Bluetooth Supp...		0.2%	7.1 MB	0 MB/s	0 M
Desktop Window Manager		0.2%	70.8 MB	0 MB/s	0 M
Service Host: DCOM Server Pr...		0.2%	11.0 MB	0 MB/s	0 M
Service Host: Remote Procedu...		0.1%	11.9 MB	0 MB/s	0 M
Service Host: Device Associati...		0.1%	1.8 MB	0 MB/s	0 M
Service Host: Network Connec...		0.1%	1.2 MB	0 MB/s	0 M
Windows Explorer		0.1%	179.1 MB	0 MB/s	0 M
Windows Terminal Host		0.1%	53.8 MB	0 MB/s	0 M
git		0%	1.0 MB	0 MB/s	0 M
conhost		0%	0.8 MB	0 MB/s	0 M
git		0%	0.9 MB	0 MB/s	0 M
Node.js JavaScript Runtime		0%	138.5 MB	0.4 MB/s	0 M

12.2 Stress evidence

Windows PowerShell

```

PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/stop-backend.ps1
Stopped backend PID 23504.
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/start-backend.ps1 -Scenario EvidenceStress
Backend PID 30872 ready at http://127.0.0.1:3000; database reseeded on startup.
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/run-jmeter.ps1 -Scenario Stress
Completed Stress: threads=400 ramp=60s duration=180s
JTL: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Stress-20260903201249\23127124_Stress_20260903.jtl
HTML: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Stress-20260903201249\html
Resources: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Stress-20260903201249\resource-usage.csv
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5>

```

Task Manager

Type a name, publisher,...

Processes

Run new task End task Efficiency mode

Name	Status	22% CPU	89% Memory	6% Disk	0% Network
Microsoft Edge (73)		0%	1,801.6 MB	0 MB/s	0.1 M
Codex (10)		4.9%	1,028.6 MB	0.1 MB/s	0 M
Microsoft OneDrive		6.2%	762.0 MB	0.1 MB/s	0 M
Lark (9)		0.5%	306.0 MB	0.1 MB/s	0.1 M
Antimalware Service Executable		1.2%	261.3 MB	0.1 MB/s	0 M
Firefox (11)		0%	232.5 MB	0.1 MB/s	0 M
Spotify (7)		0.5%	185.4 MB	0 MB/s	0 M
Windows Explorer		0%	175.5 MB	0 MB/s	0 M
Node.js JavaScript Runtime		0%	163.4 MB	0 MB/s	0 M
Search (7)		0%	136.9 MB	0.1 MB/s	0 M
codex		0.5%	100.5 MB	0.1 MB/s	0 M
Desktop Window Manager		0.1%	89.5 MB	0 MB/s	0 M
Task Manager		0.8%	79.4 MB	0 MB/s	0 M
Microsoft OneDrive Sync Serv...		5.7%	69.9 MB	0.7 MB/s	0 M
Secure System		0%	67.5 MB	0 MB/s	0 M
Windows Terminal Host		0%	54.0 MB	0 MB/s	0 M
Service Host: Diagnostic Polic...		0%	36.4 MB	0.1 MB/s	0 M
Start		0%	35.5 MB	0 MB/s	0 M
Windows PowerShell		0%	24.5 MB	0 MB/s	0 M
Service Host: UtcSvc		0%	22.3 MB	0 MB/s	0 M
Mobile devices (2)		0%	17.9 MB	0 MB/s	0 M

Search

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12.3 Spike evidence

Windows PowerShell

```

PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/stop-backend.ps1
Stopped backend PID 30872.
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/start-backend.ps1 -Scenario EvidenceSpike
Backend PID 27784 ready at http://127.0.0.1:3000; database reseeded on startup.
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5> powershell -ExecutionPolicy Bypass -File scripts/run-jmeter.ps1 -Scenario Spike
Completed Spike: threads=600 ramp=2s duration=90s
JTL: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Spike-20260903201701\23127124_Spike_20260903.jtl
HTML: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Spike-20260903201701\html
Resources: C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5\results\Spike-20260903201701\resource-usage.csv
PS C:\Users\thoai\OneDrive\Documents\ChatGPT\Homework 5>

```

Task Manager

Type a name, publisher,...

Processes

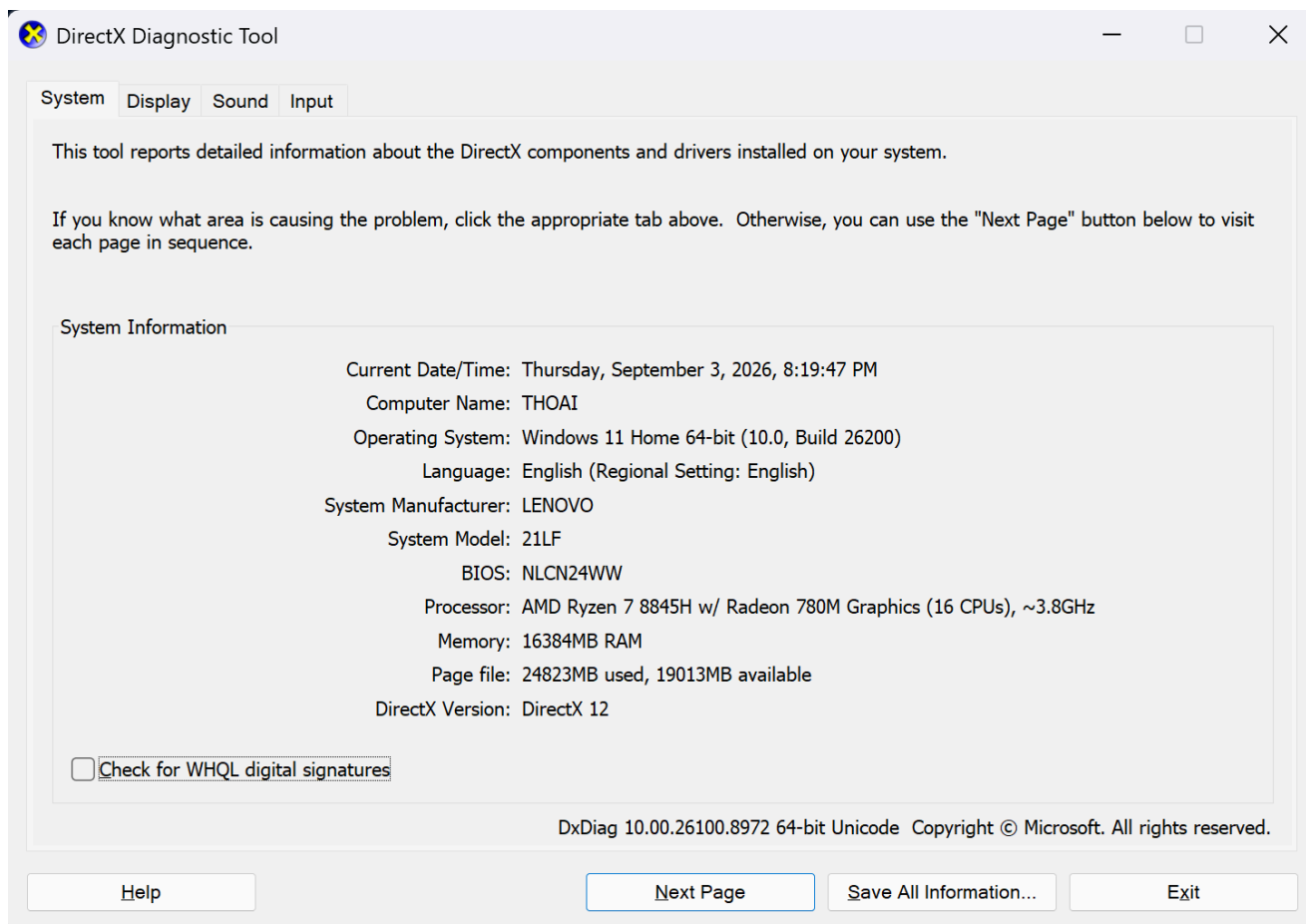
Run new task End task Efficiency mode

Name	Status	20% CPU	89% Memory	4% Disk	0% Network
Microsoft OneDrive Sync Serv...		7.5%	69.3 MB	0.4 MB/s	0 M
Microsoft OneDrive		6.1%	899.3 MB	0.1 MB/s	0 M
Task Manager		1.3%	78.9 MB	0 MB/s	0 M
Desktop Window Manager		0.8%	86.3 MB	0.1 MB/s	0 M
System		0.7%	0.1 MB	0.4 MB/s	0 M
Antimalware Service Executable		0.6%	205.8 MB	0.1 MB/s	0 M
Spotify (7)		0.6%	183.5 MB	0.3 MB/s	0 M
Codex (10)		0.5%	986.9 MB	0.1 MB/s	0 M
Service Host: Bluetooth Supp...		0.3%	7.4 MB	0 MB/s	0 M
Windows Explorer		0.3%	176.9 MB	0.1 MB/s	0 M
Mobile devices (2)		0.3%	18.4 MB	0 MB/s	0 M
Service Host: Remote Procedu...		0.2%	12.3 MB	0 MB/s	0 M
Service Host: Network Service		0.2%	4.9 MB	0 MB/s	0 M
Windows Driver Foundation - ...		0.2%	1.0 MB	0 MB/s	0 M
Service Host: Windows Health...		0.2%	8.3 MB	0 MB/s	0 M
Windows Terminal Host		0.2%	53.9 MB	0 MB/s	0 M
Lark (9)		0.1%	295.9 MB	0.1 MB/s	0 M
Service Host: Clipboard User S...		0.1%	2.2 MB	0 MB/s	0 M
SnippingTool		0%	32.8 MB	0 MB/s	0 M
Microsoft Edge (73)		0%	1,778.5 MB	0 MB/s	0.1 M
Client Server Runtime Process		0%	1.3 MR	0 MR/s	0 M

Search

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12.4 Hardware evidence



13. Demo video

The student supplied the narrated demonstration as an unlisted YouTube video:

<https://youtu.be/kmkiT8y2rXQ>. The final recorded demo workload completed 5,831 HTTP samples with zero errors, 126.95 RPS, p95 of 7 ms, and a maximum backend working set of 88.36 MB.

Appendix A - AI Audit declaration

I use AI tools for the following tasks. The interaction log records the AI tool, date/time, student prompt, output, corrections, and generated artifacts. The full audit is supplied separately in Markdown and PDF. Calibration failures are retained rather than hidden because they demonstrate the required human review.